

# Workday Data Query and Reporting Agent

Natural-language-to-SQL agent for ad hoc Workday-style HR reporting

## BUSINESS SCENARIO

HR and IT leaders using a Workday-style HR system need quick answers to ad hoc questions, such as headcount on leave by region this month or average time-to-fill for open roles, but must currently submit requests to a report-writing team or learn a report-builder tool themselves. This creates multi-day delays for questions that should take minutes to answer, and leaders often give up and rely on stale, previously generated reports instead.

## EXPECTED SOLUTION

An agent that accepts plain-English questions, retrieves the relevant schema and data-dictionary context via RAG, generates a validated SQL query against the HR database, executes it, and returns a clear natural-language answer with a supporting table. A faithfulness self-check compares the generated answer against the actual query results before responding, and the agent refuses or asks for clarification on ambiguous or out-of-scope requests rather than guessing at a schema mapping.

## KEY CHALLENGES TO SOLVE

- Preventing the agent from generating SQL against tables or columns that do not exist in the real schema.
- Catching cases where the generated query runs successfully but answers a subtly different question than what was asked.
- Supporting natural follow-up drill-downs, such as "now break that down by department", without regenerating the whole query from scratch.

## EVALUATION RUBRIC

| # | CRITERION                 | WEIGHT | WHAT IS MEASURED                                                                                                         |
|---|---------------------------|--------|--------------------------------------------------------------------------------------------------------------------------|
| 1 | SQL Generation Accuracy   | 40%    | Generated query returns correct results, verified against a ground-truth query set.                                      |
| 2 | Answer Faithfulness       | 35%    | RAGAS-style check that the final natural-language answer reflects the actual query results, not invented figures.        |
| 3 | Schema-Grounding & Safety | 25%    | Agent references only real schema retrieved via RAG and rejects out-of-scope requests, tested with a red-team query set. |

## EXPECTED WORKFLOW

1. Natural-language question **SEQ** schema RAG node retrieves relevant tables/columns from the data dictionary
2. SQL generation node drafts a query constrained to the retrieved schema
3. SQL validation and execution node (read-only DB tool call) runs the query safely
4. Result-to-answer node drafts the response, then runs a faithfulness self-check against the raw results
5. Answer and supporting table returned to the frontend; ambiguous asks trigger a clarifying question instead

## EXPECTED STACK

**API:** FastAPI

**Frontend:** React or HTML dashboard with a results table and query-history view

**Database:** SQL database seeded with a Workday-style HR schema and sample data

**RAG:** Retrieval over schema/data-dictionary docs, evaluated with RAGAS

**Agent:** LangGraph, sequential plan-generate-validate-answer flow

**Tools:** Read-only SQL execution tool, schema/metadata lookup tool

**Context:** Conversation memory so follow-up drill-down questions reuse the prior query context